Supplementary Material

Clean-Teacher Distillation on Corrupted Sensor Inputs for Robust Multimodal Semantic Segmentation

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This document complements the main paper with tables that are too detailed for the main text. Table and figure numbers without the prefix S refer to the main paper. All numbers of our model are the mean over the training runs stated in the main paper unless a caption says otherwise.

1 Missing sensors per subset on DELIVER

Table S1 expands the entire-missing average of Table 5 (benchmark protocol) into the 15 subsets of removed sensors, measured with the public benchmark code on the validation set. The released CMNeXt weights reproduce the benchmark's reported values, and the averages in the last rows are those of Table 5. CMNeXt collapses whenever depth is removed (22.9 with depth alone missing, 3.2 or less with depth and RGB missing), whereas our model keeps at least 24.9 in every subset and matches or exceeds its clean score when event, LiDAR, or both are removed (71.5 to 72.0 against 71.3).

2 MUSES test results per condition and class

Table S2 gives the official test scores of our three MUSES models per condition, next to the per-condition results that MM SAM-adapter reports for the same two-sensor settings. Table S3 gives the per-class scores of the same submissions. Each of our entries is one submission (the run with the higher validation score), so differences of about one point between models or conditions are within what a different training run could change.

3 MCubeS per class

Table S4 compares, per material class, our full model with the clean-trained model without the high-resolution path that serves as its teacher.

4 Dependence on depth per condition

Table S5 lists the per-condition drops plotted in Fig. 7 of the main paper.

5 Planned for revision: C2D2 on other fusion architectures

[TODO: Only if reviewers ask (lab-plan DRN-261029-02/03, revision reserve). Apply the full C2D2 scheme (clean pass, degraded pass, frozen clean teacher) to DGFusion and CAFuser and report clean mIoU, EMM average, and the per-condition depth drop next to their single-pass re-training of Table S5 and of the re-trained comparison in the main paper.]

References

[1] C. Liao, K. Lei, X. Zheng, J. Moon, Z. Wang, Y. Wang, D. P. Paudel, L. Van Gool, and X. Hu, "Benchmarking multi-modal semantic segmentation under sensor failures: Missing and noisy modality robustness," *arXiv preprint arXiv:2503.18445*, 2025.

[2] I. Curti, P. Zama Ramirez, A. Petrelli, and L. Di Stefano, "Multimodal SAM-adapter for semantic segmentation," *IEEE Access*, 2025, arXiv:2509.10408.

Table S1: Entire-missing evaluation on the DELIVER validation set per removed-sensor subset (mIoU, %), public benchmark code [1]. R: RGB, D: depth, E: event, L: LiDAR.

Removed sensors	CMNeXt (released)	Ours (mean of 3 runs)
none (clean)	66.33	**71.34**
R	50.98	**62.28**
D	22.92	**63.13**
E	66.38	**71.53**
L	66.27	**71.62**
R + D	3.15	**36.93**
R + E	50.83	**60.71**
R + L	50.80	**62.23**
D + E	22.50	**63.33**
D + L	22.91	**63.24**
E + L	66.33	**71.97**
R + D + E	2.86	**24.93**
R + D + L	3.16	**31.29**
R + E + L	50.59	**60.25**
D + E + L	22.49	**62.98**
Average over 15 subsets	37.90	**58.52**
E(.2)	54.46	**67.15**
E(.1)	60.41	**69.46**
E(.05)	63.38	**70.45**

Table S2: MUSES test set per condition (mIoU, %, official evaluation server). MM SAM-adapter values are from [2] (Table 6); ours are one submission each.

Model	All	Day	Night	Clear	Fog	Rain	Snow
MM SAM-adapter, RGB-L	**81.07**	83.34	74.97	80.82	74.12	**80.00**	**80.85**
MM SAM-adapter, RGB-E	79.92	**83.39**	72.38	**81.52**	68.97	78.92	78.72
Ours, RGB-L	80.46	81.53	**77.56**	80.55	74.00	79.15	80.07
Ours, RGB-E	79.91	82.68	74.27	80.90	72.52	78.60	77.91
Ours, RGB-L-R-E	80.01	81.97	75.83	79.86	**79.23**	79.34	78.84

Model	Clear day	Clear night	Fog day	Fog night	Rain day	Rain night	Snow day	Snow night
Ours, RGB-L	81.66	**77.04**	72.85	67.84	78.86	**74.36**	72.06	**78.16**
Ours, RGB-E	**83.15**	74.45	74.33	64.41	79.83	72.65	**72.72**	72.00
Ours, RGB-L-R-E	82.02	74.81	**80.03**	**72.54**	**80.38**	74.14	72.69	77.12

Table S3: MUSES test set per class (IoU, %, official evaluation server, all conditions).

Class	RGB-L	RGB-E	RGB-L-R-E
Road	**97.23**	97.17	97.09
Sidewalk	**87.26**	87.14	86.99
Building	**93.59**	93.01	93.55
Wall	82.04	78.73	**82.42**
Fence	**67.90**	64.12	66.26
Pole	65.18	**65.58**	65.34
Traffic light	**71.93**	71.56	71.61
Traffic sign	75.18	73.00	**75.27**
Vegetation	**89.87**	88.57	89.70
Terrain	**79.13**	76.43	77.74
Sky	96.89	96.52	**97.10**
Person	72.92	**73.11**	72.79
Rider	59.51	**62.28**	60.05
Car	**94.64**	94.38	94.35
Truck	**77.64**	77.63	77.33
Bus	95.38	95.70	**95.77**
Train	93.96	**94.43**	93.12
Motorcycle	**59.22**	58.81	56.74
Bicycle	69.26	**70.19**	67.02
mIoU	**80.46**	79.91	80.01

Table S4: MCubeS test set per class (IoU, %, mean of two runs): the clean-trained model without the high-resolution path and our full model.

Class	Clean-trained, no high-res.	Ours
Asphalt	88.81	**89.62**
Concrete	**55.99**	55.80
Metal	64.56	**65.10**
Road marking	74.08	**77.13**
Fabric	**45.16**	42.01
Glass	61.72	**61.84**
Plaster	0.19	**1.51**
Plastic	50.53	**52.39**
Rubber	27.18	**32.89**
Sand	61.97	**64.91**
Gravel	67.91	**72.28**
Ceramic	40.21	**42.30**
Cobblestone	73.47	**76.41**
Brick	**52.65**	51.27
Grass	**62.72**	60.90
Wood	**53.03**	50.52
Leaf	**77.78**	77.25
Water	85.30	**86.31**
Human	35.98	**38.40**
Sky	96.21	**96.67**
mIoU	58.78	**59.78**

Table S5: Drop in mIoU (points) on the DELIVER test set when depth is removed (zero-filled), per condition; lower is better. Data of Fig. 7 of the main paper.

Model	Sun	Cloud	Rain	Fog	Night	Low exp.
DGFusion (our reproduction)	23.1	24.9	27.6	25.9	32.6	37.1
Ours, clean training, no high-res. path (2 runs)	14.6	14.8	14.3	14.4	17.9	25.0
Ours (3 runs)	1.3	1.4	1.5	0.6	4.3	9.6